

EDUCATION

Heritage High School, Frisco, TX

Class of 2027

GPA: 4.00 UW / 5.361 W · Rank 11/481 · ACT 36 · Nine AP scores of 5 · AP Scholar with Distinction

Senior Coursework: AP Calculus BC, AP Physics 2, Computer Science III, Mobile App Programming

UT Dallas NLP with Python Workshop, Summer 2025 — Implemented rule-based text simplification using dependency parsing, frequency-based lexical substitution, clause splitting, and passive-to-active transformation

RESEARCH

Predicting Cognitive Function Using Transformer-Derived Speech Representations and Longitudinal Coherence Features[↗] — First Author

2025–Present

Mentor: Joe Xiao (PhD student) · [medRxiv preprint \(Aug. 2026\)](#) · Under review, *Frontiers in Aging Neuroscience*

- Formulated next-visit MMSE forecasting as supervised regression on DementiaBank Pitt Corpus (173 longitudinal prediction samples, 126 patients); extracted 12 per-visit linguistic features including POS-based fragment ratio, named-entity rate, 50-token MATTR, filler rate, and perseveration.
- Engineered Sentence-BERT discourse representations using adjacent-sentence cosine similarity, global coherence, semantic slope, topic drift, embedding variance, visit-to-visit Δ features, and temporal metadata.
- Built a 53-dimensional feature representation combining 20 PCA components of 384-dimensional SBERT embeddings with 9 clinical/demographic covariates; applied RobustScaler, correlation pruning, permutation-importance selection, and Optuna/TPE optimization of LightGBM across 200 trials.
- Diagnosed patient-identity leakage and rebuilt evaluation with patient-grouped 5-fold GroupKFold, retaining augmented copies within patient folds and scoring original samples only: R^2 0.840 $\pm$ 0.015, RMSE 2.75, MAE 2.02, Pearson r 0.917.
- Controlled feature-source ablation showed clinical-only R^2 0.72, SBERT-only 0.46, linguistic-only 0.12, and clinical+SBERT 0.76; reported non-nested feature selection/tuning as an optimism limitation.

X-Step: Four-Site Plantar-Pressure Insole[↗] — ML Lead, 2nd of 5 Authors

2025–Present

UC Santa Cruz AIEA Lab (Prof. Leilani Gilpin) · MIT CSAIL co-authors · Under review, *IEEE-EMBS IECBES 2026*

- Implemented a 59-feature plantar-pressure representation from four FSR402 sensors sampled at 25 Hz over BLE from an ESP32, including per-site peak/mean/CV, pressure-time integral, cadence, stance ratio, and anterior load share, with a 540-parameter logistic-regression head.
- Designed a sparse-vs-dense sensing study on 149 recordings from 15 adults using a public 32-cell insole: four sites preserved waveform shape (median r = 0.72–0.99) but not magnitude (peak-to-peak r = 0.43–0.74); center-of-pressure tracking preserved anteroposterior motion (r = 0.81) but not mediolateral motion (r = 0.01).
- Quantified subject-identity leakage on a synthetic 24-subject cohort: macro-F1 0.885 under subject-grouped evaluation vs. 0.931 ungrouped; documented the resulting generalization constraint.

COMPUTING EDUCATION

KidCode[↗] — Founder & Lead Instructor · Frisco, TX

2024–Present

- Designed a 3-year, 120-class-day, 20-project Python/NLP curriculum spanning control flow, file I/O, JSON, exceptions, OOP/inheritance, assert-based unit testing, and spaCy NLP: tokenization, lemmatization, POS tagging, sentence segmentation, NER, Matcher patterns, and rule-based chatbots.
- Taught 124 students from 10 Frisco neighborhoods across three cohorts (35/53/36); designed 13 curriculum-aligned pre/post assessment pairs. Paired results ($n=104$): 41.7% $\rightarrow$ 58.8% (+17.1 pp; 99/104 improved); no comparison group.
- Students completed 1,214 working programs across the 20 projects, 572 of them using spaCy: deeptanshudevatha.com/projects/kidcode.

Ayusha Nilayam[↗] — Programming & Math Instructor · Hyderabad, India (remote)

Nov. 2025–Jul. 2026

- Taught 38 weekly Zoom sessions of Python and mathematics to 11 children at a Hyderabad children's home, adapting instruction around one shared laptop, unreliable internet, and a Telugu–English language gap; taught in person for two weeks.

ENGINEERING

X-Step[↗] — Co-Founder & ML/Mobile Lead

2025–Present

- Engineered a four-site plantar-pressure insole prototype using ESP32 + FSR402 sensors, 25 Hz BLE acquisition, temporal windowing, biomechanical feature extraction, and real-time gait visualization/inference through a React Native mobile application.
- Integrated hardware sensing, ML inference, and mobile software into an end-to-end prototype; led ML and mobile development across the product stack.
- Team: Diamond Challenge Semifinalist; [Texas A&M Ideas Challenge Finalist](#).

ShopAI[↗] — Creator & Sole Developer

2025–Present

- Architected a React Native/Expo + Express/TypeScript system on AWS EC2 behind Nginx with a [retailer-adaptor abstraction](#) normalizing nine chains across OAuth2 REST, cookie-session GraphQL, Playwright, and server-rendered HTML integrations.
- Engineered concurrent fan-out search, per-store failure states, query-correction/relevance/ranking pipelines, and a per-ZIP authentication/session warm-up state machine; live on the [Apple App Store](#).
- Implemented a deterministic optimizer exhaustively enumerating 127 non-empty store subsets, producing cost-, drive-time-, stop-count-, and weighted multi-criteria plans; planning/search path makes zero LLM calls.
- Prototype predecessor: [TechLit Hackathon 1st](#) · [Hack4Humanity 3rd](#) · [PeddieHacks Innovation Prize](#).

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, SQL

ML/NLP: LightGBM, scikit-learn, Optuna, Sentence-BERT, spaCy, PCA, GroupKFold, permutation importance, PyTorch

Systems: React Native/Expo, Node.js/Express, REST, GraphQL, OAuth2, Playwright, AWS EC2, Nginx · Embedded: ESP32, BLE, FSR sensing